FinSight AI

AI-Based Financial Identification and Forecasting

Product documentation | Version 1.0 | Local-first Flask application

FinSight AI analyzes an individual's demographic and financial information to produce a financial profile, a risk assessment, a future economic forecast, and explainable ML insights with visualizations and actionable, non-prescriptive guidance.

1. Product overview

FinSight AI is a standalone commercial-style web application. A user completes a financial assessment; the system validates the input, engineers interpretable features, runs a trained classifier, explains the prediction with SHAP, shows an economic forecast, and optionally compares what-if scenarios side-by-side.

2. Problem statement

Individuals lack clear, data-driven feedback on their financial position and on which factors matter most. FinSight AI converts factors such as age, occupation, education, income signals, location, and economic indicators into an interpretable profile with honest, model-based insights.

3. Key features

- Financial assessment with full server-side validation.
- Financial profile classification (Emerging / Developing / Established) with confidence and Low / Moderate / High risk bands.
- Four-model comparison (Logistic Regression, Random Forest, Gradient Boosting, XGBoost) with best-model selection by ROC-AUC.
- SHAP global importance plus live per-person local explanations.
- Holt-Winters 12-month economic forecast with intervals and trend interpretation.
- What-if scenario comparison (baseline vs scenario, labelled as simulation).
- Stratified 5-fold validation, leakage-safe pipelines, statistical summaries.

4. User workflow

- Dashboard: product summary, model snapshot, Start Financial Assessment CTA.
- Assessment: enter age, employment type, education, marital status, occupation, household relationship, race, sex, country, investment gains/losses, hours per week.
- Result: profile, risk band, strength score, SHAP drivers, insights, forecast mini-chart, inputs.
- Explainability: global SHAP bar + beeswarm + ranked table + reading guide.
- Forecast: history, 12-month forecast, intervals, values table.
- What-If: baseline + scenario overrides, side-by-side comparison with delta.
- Performance: real model comparison, confusion matrix, ROC, validation, statistics.

5. Technology stack

Python, pandas, NumPy, scikit-learn, XGBoost, SHAP, statsmodels, Flask, Jinja2 templates, HTML/CSS/vanilla JavaScript, Plotly.js charts, matplotlib (training plots), joblib persistence, pytest. Lightweight: runs locally on an ordinary Windows laptop with no GPU.

6. Architecture summary

data ingestion -> preprocessing -> feature engineering -> model training -> evaluation -> persisted model artifacts -> Flask prediction API -> dashboard frontend. Training is offline (scripts/train.py); Flask loads persisted artifacts at startup so predictions are fast. Modules: app/__init__.py, app/routes.py, app/services/predictor.py, forecaster.py, whatif.py, app/utils/validation.py, scripts/feature_engineering.py, models.py, forecast.py, train.py, evaluate.py.

7. ML pipeline

Load Adult data -> schema validation -> cleaning (? to missing, numeric coercion) -> derived features (net-capital, log scales, bands, tiers, region, employment-stability, capital-position) -> stratified 80/20 split -> 4 candidate pipelines (scaling only for Logistic Regression) -> 5-fold stratified CV (ROC-AUC) -> test evaluation -> best-model selection -> SHAP artifacts -> confusion matrix / ROC plots -> persist model + preprocessor -> metrics, validation, stats, forecast artifacts. random_state=42 throughout.

8. Dataset and preprocessing

Classification dataset: UCI Adult / Census Income (public); source: https://archive.ics.uci.edu/dataset/20/census+income; rows: 48842. Target: income >50K vs <=50K adapted as a financial-profile signal (probability of >50K = financial-strength score; NOT a credit score or approval). Missing values: median imputation (numeric) and 'Unknown' constant (categorical) inside the pipeline. Forecasting uses a separate monthly Household Financial Conditions Index series (documented synthetic illustrative series, 132 months 2015-2025) because the classification data is cross-sectional. See data/README.md for full provenance.

9. Model comparison (actual results)

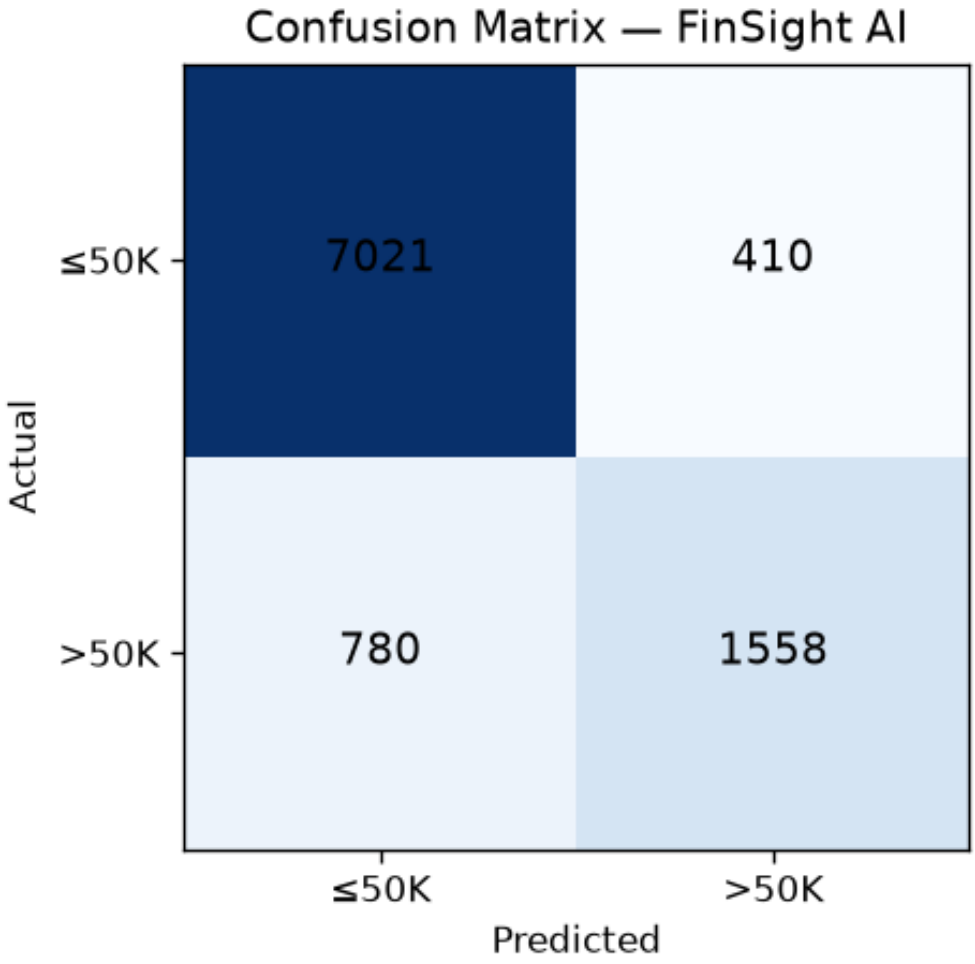
logistic_regression: accuracy=0.8599, precision=0.7568, recall=0.6108, F1=0.676, ROC-AUC=0.914, CV AUC=0.9133 +/- 0.0013

random_forest: accuracy=0.8597, precision=0.7402, recall=0.6373, F1=0.6849, ROC-AUC=0.91, CV AUC=0.9079 +/- 0.0016

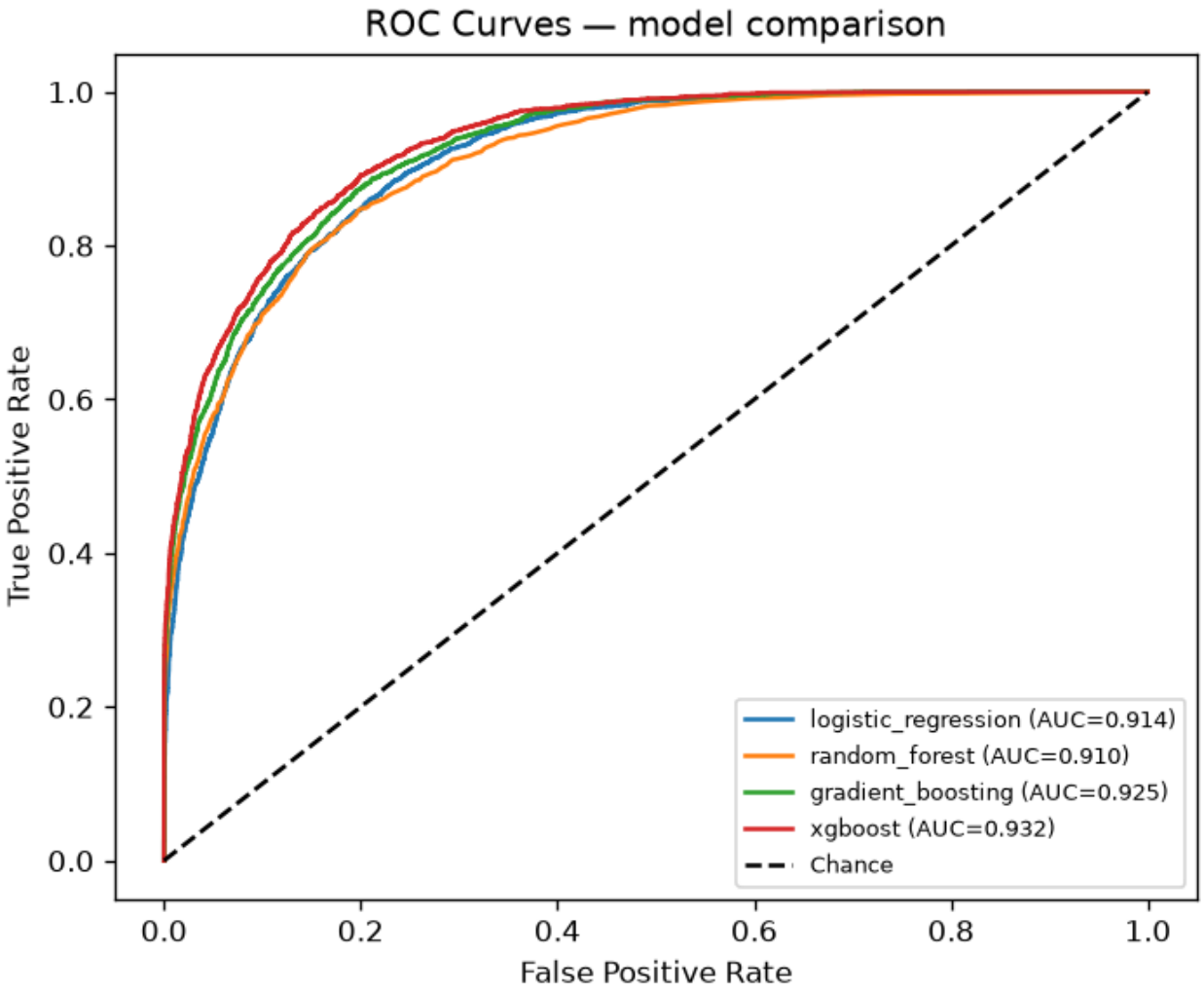
gradient_boosting: accuracy=0.8697, precision=0.7915, recall=0.6185, F1=0.6944, ROC-AUC=0.9247, CV AUC=0.9204 +/- 0.0022

xgboost [BEST]: accuracy=0.8782, precision=0.7917, recall=0.6664, F1=0.7236, ROC-AUC=0.9319, CV AUC=0.9293 +/- 0.0011

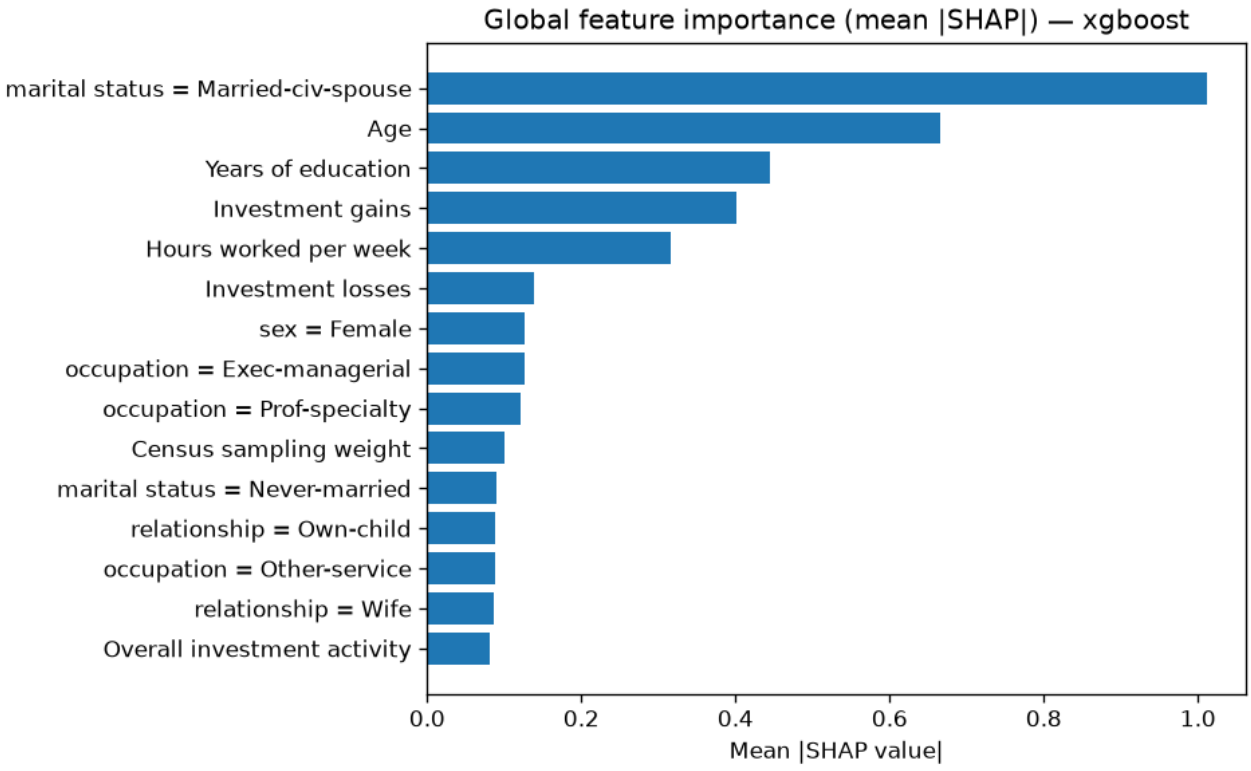
confusion_matrix.png



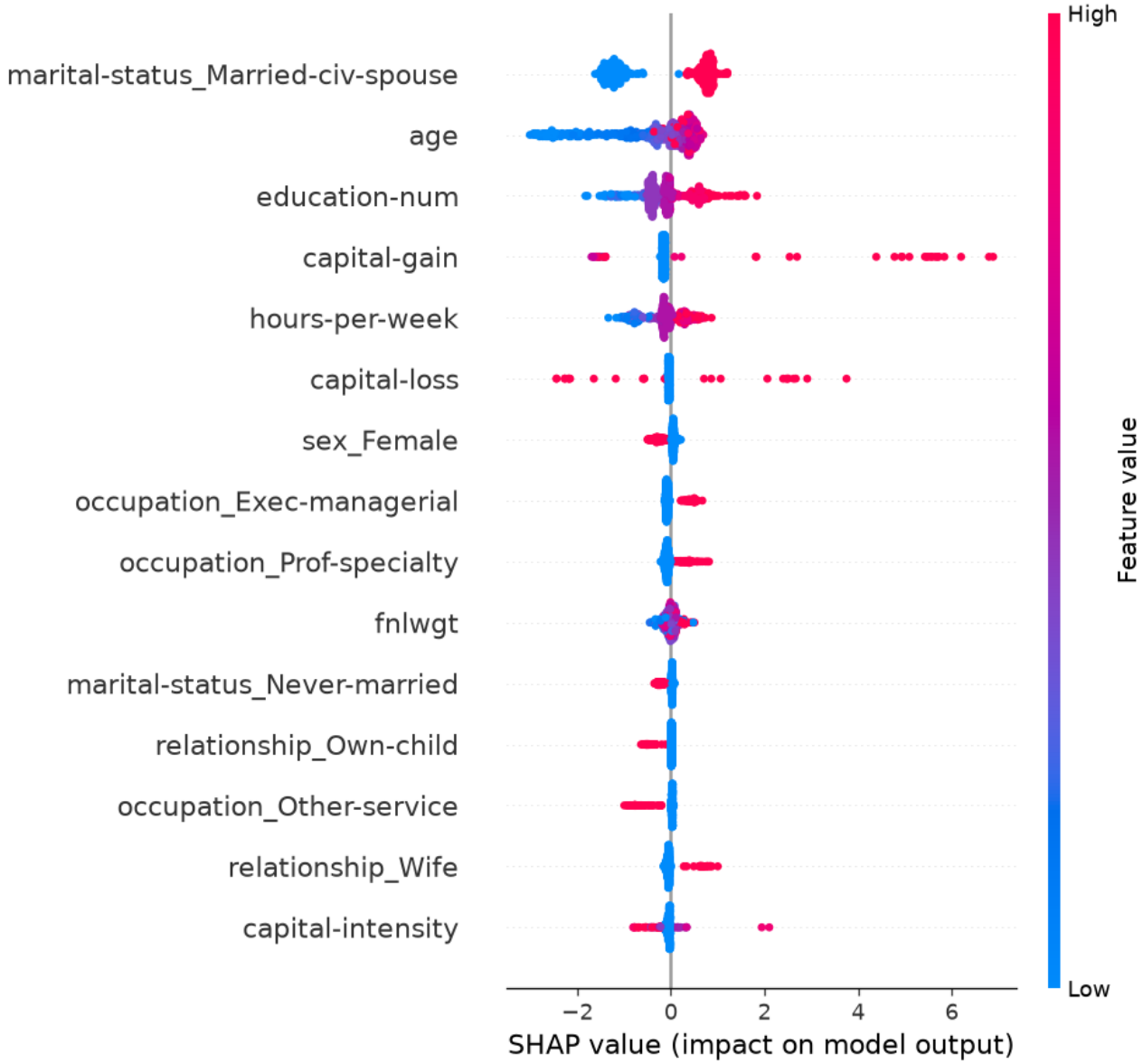
roc_curve.png


ROC Curves — model comparison
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
logistic_regression (AUC=0.914)
random_forest (AUC=0.910)
gradient_boosting (AUC=0.925)
xgboost (AUC=0.932)
Chance

shap_bar.png


Global feature importance (mean |SHAP|) — xgboost
marital status = Married-civ-spouse
Age
Years of education
Investment gains
Hours worked per week
Investment losses
sex = Female
occupation = Exec-managerial
occupation = Prof-specialty
Census sampling weight
marital status = Never-married
relationship = Own-child
occupation = Other-service
relationship = Wife
Overall investment activity
0.0
0.2
0.4
0.6
0.8
1.0
Mean |SHAP value|

shap_beeswarm.png



10. Explainability / SHAP

SHAP TreeExplainer (tree-path-dependent) on the selected tree model. Global importance is precomputed over a 500-row sample; local contributions are computed live per prediction with positive (strengthening) vs negative (weakening) contributors, values, and plain-language sentences. The UI and this document state that SHAP describes learned associations, not causation.

- marital status = Married-civ-spouse: mean|SHAP| = 1.0116
- Age: mean|SHAP| = 0.66528
- Years of education: mean|SHAP| = 0.44447
- Investment gains: mean|SHAP| = 0.40074
- Hours worked per week: mean|SHAP| = 0.31528
- Investment losses: mean|SHAP| = 0.13848
- sex = Female: mean|SHAP| = 0.12635
- occupation = Exec-managerial: mean|SHAP| = 0.12562
- occupation = Prof-specialty: mean|SHAP| = 0.1204

- Census sampling weight: mean|SHAP| = 0.10067

11. Forecasting

Method: Holt-Winters Exponential Smoothing (additive trend + seasonality). Horizon: 12 months. Trend: upward. The index is projected to move upward over the next 12 months (+3.46% from 141.7 to 146.6). This is a statistical extrapolation of past patterns, not a guarantee of any individual's future finances. Residual std: 1.659.

12. What-if scenario analysis

Users override gains, losses, hours, and/or education; the persisted model re-runs on baseline and scenario; the UI reports both scores, the delta, profile/risk changes, and changed fields. Labelled everywhere as a scenario simulation, not a causal prediction.

13. Validation methodology

Protocol: Stratified 80/20 train-test split + 5-fold stratified cross-validation (n_splits=5, random_state=42). Why: The dataset is cross-sectional with one row per person; no repeated individuals exist, so entity-aware GroupKFold is unnecessary and stratified KFold preserves class ratios. Preprocessing is inside each pipeline to prevent leakage. Temporal validation is not applicable because the classification data has no genuine time dimension; forecasting is validated separately on its own time series. Train size: 39073, test size: 9769.

Statistical associations (complementary, not causal):

- age: r=0.2293, logit coef=0.04443, p=0.0
- education-num: r=0.3354, logit coef=0.33742, p=0.0
- hours-per-week: r=0.2306, logit coef=0.04182, p=0.0
- net-capital: r=0.2124, logit coef=0.00024, p=0.0

14. Actual model results and metrics

Best model: xgboost. Test metrics: {"accuracy": 0.8782, "precision": 0.7917, "recall": 0.6664, "f1": 0.7236, "roc_auc": 0.9319, "cv_roc_auc_mean": 0.9293, "cv_roc_auc_std": 0.0011}. Every number above was generated by the training run in artifacts/ (metrics.json, model_comparison.json, training_summary.json); no benchmark values are invented.

15. How to run the product

Automatic (recommended): double-click run.bat (checks Python, creates .venv, installs requirements, trains if needed, starts Flask on http://localhost:5000, opens the browser). Docker (recommended if Docker Desktop is installed): docker build -t finsight-ai . and docker run --rm -p 5000:5000 finsight-ai, or double-click run-docker.bat. Manual: python -m venv .venv; activate; pip install -r requirements.txt; python scripts/train.py; python run.py; open http://localhost:5000. Full details in HOW_TO_RUN.md.

16. Limitations and responsible-use notes

- Informational estimates from historical patterns; not professional financial advice.
- No guarantee of income, returns, credit, or loan approval.
- Training data reflects historical census patterns and may encode societal biases.
- SHAP shows associations, not causation; what-if runs are simulations, not causal.
- The economic index is illustrative; forecasts extrapolate past patterns.